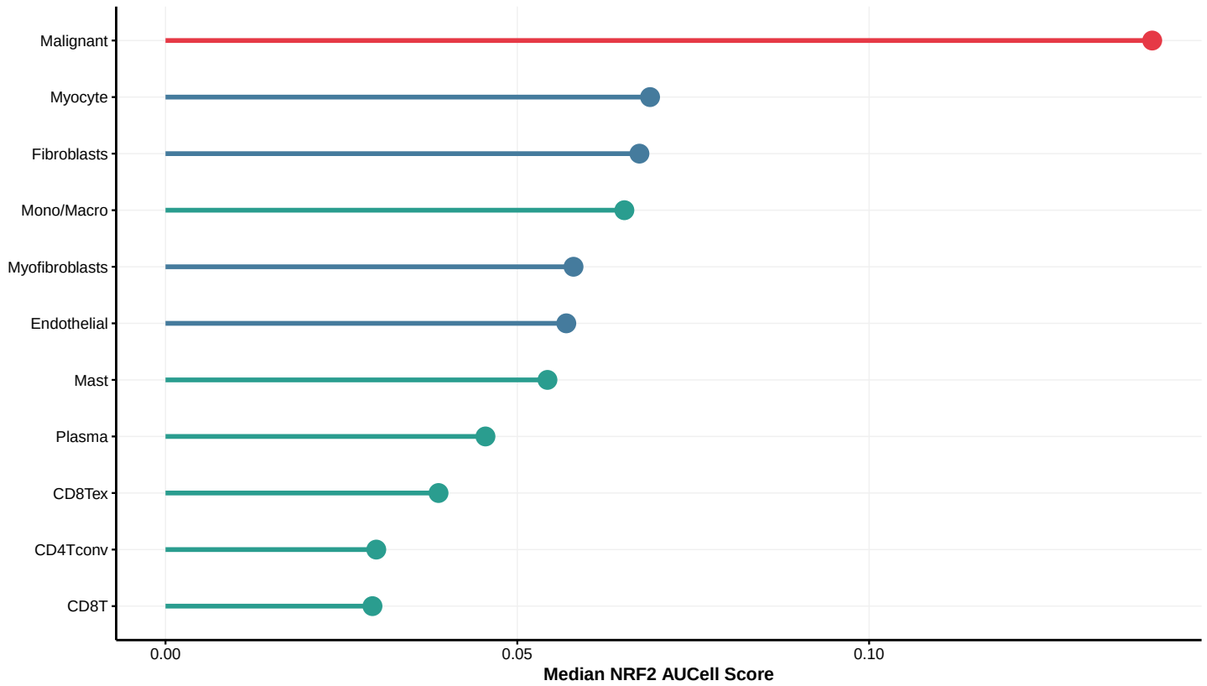
Master Single−Cell Architecture of the KEAP1/NRF2 Axis in OSCC (GSE103322 Ecosystem)

Comprehensive multi−compartment synthesis: Cell autonomy, clonal heterogeneity, 5−bridge validation, metastatic amplification & priority gene ecosystem

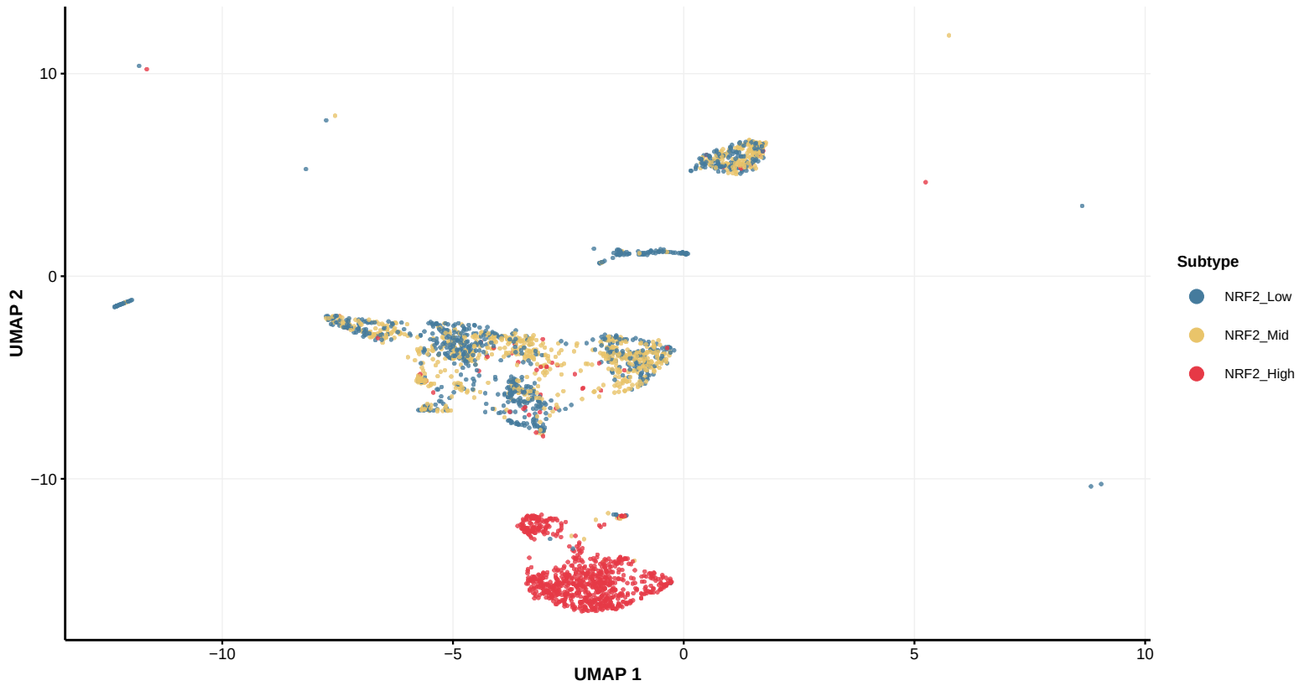
A Cellular Localisation Hierarchy

Malignant cell autonomy (Cliff's d = +0.92 vs Immune)



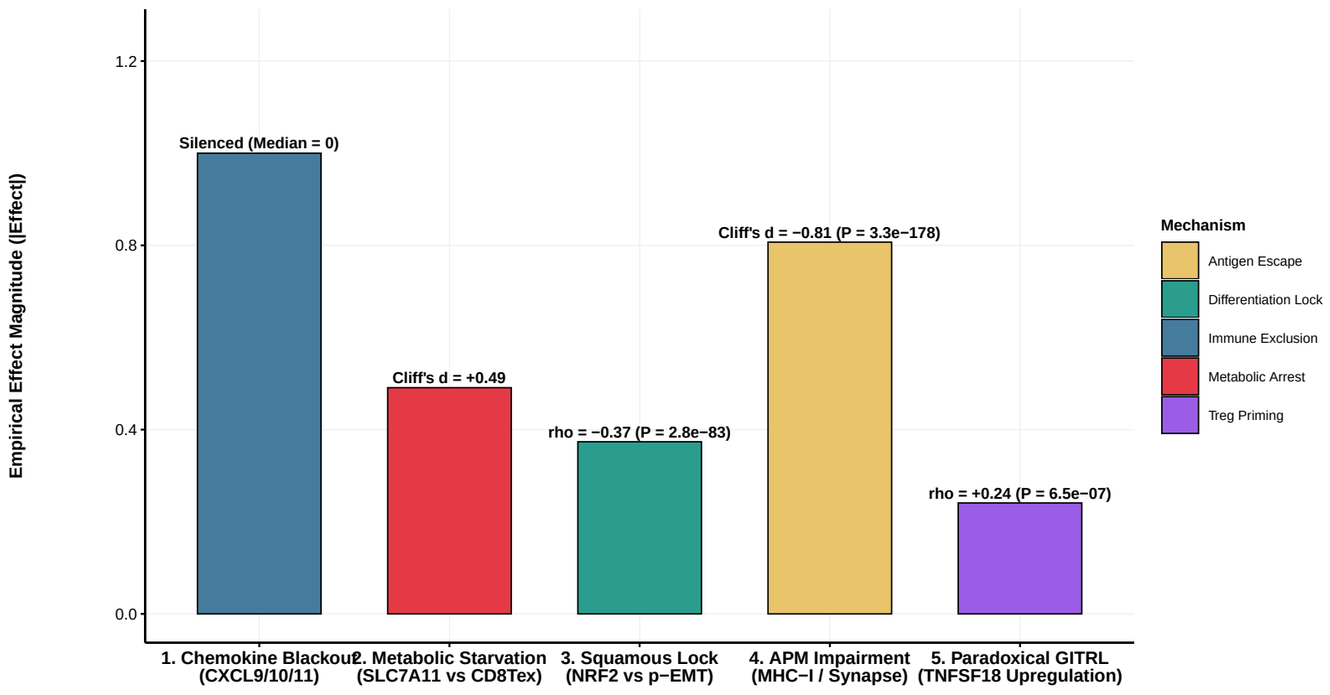
B Malignant NRF2 Clonal Topology

Focal hyperactivated clones (HNSCC20=97%, HNSCC6=91%)



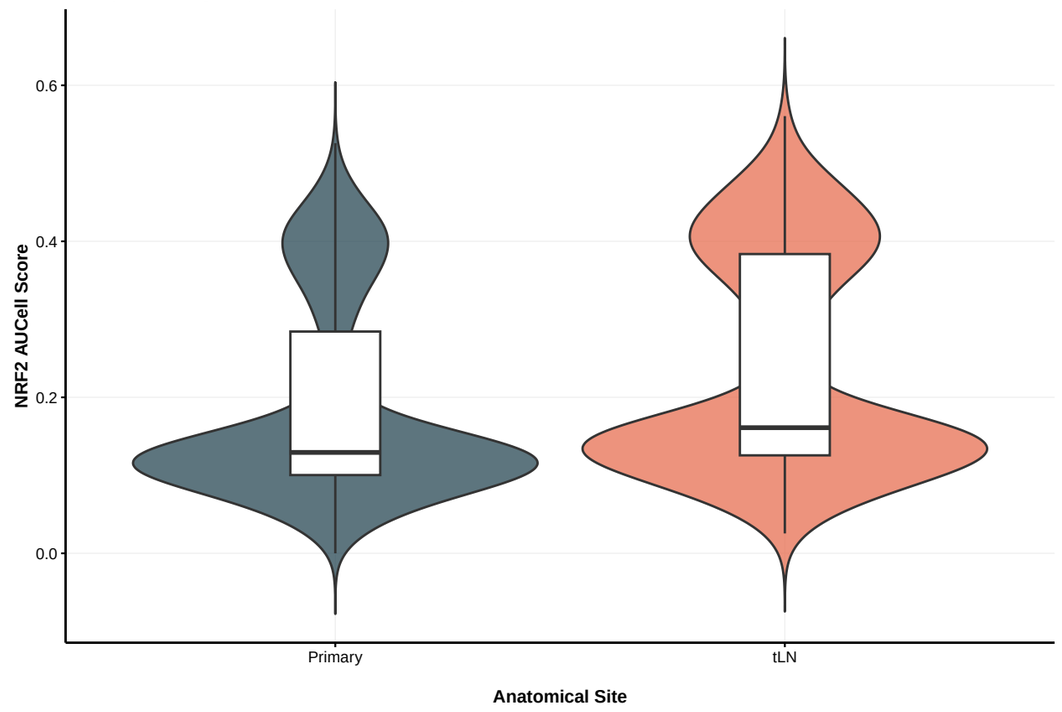
C Phase 3 Mechanistic TME Bridges Validation

Single−cell confirmation of all 5 causal desertification pillars



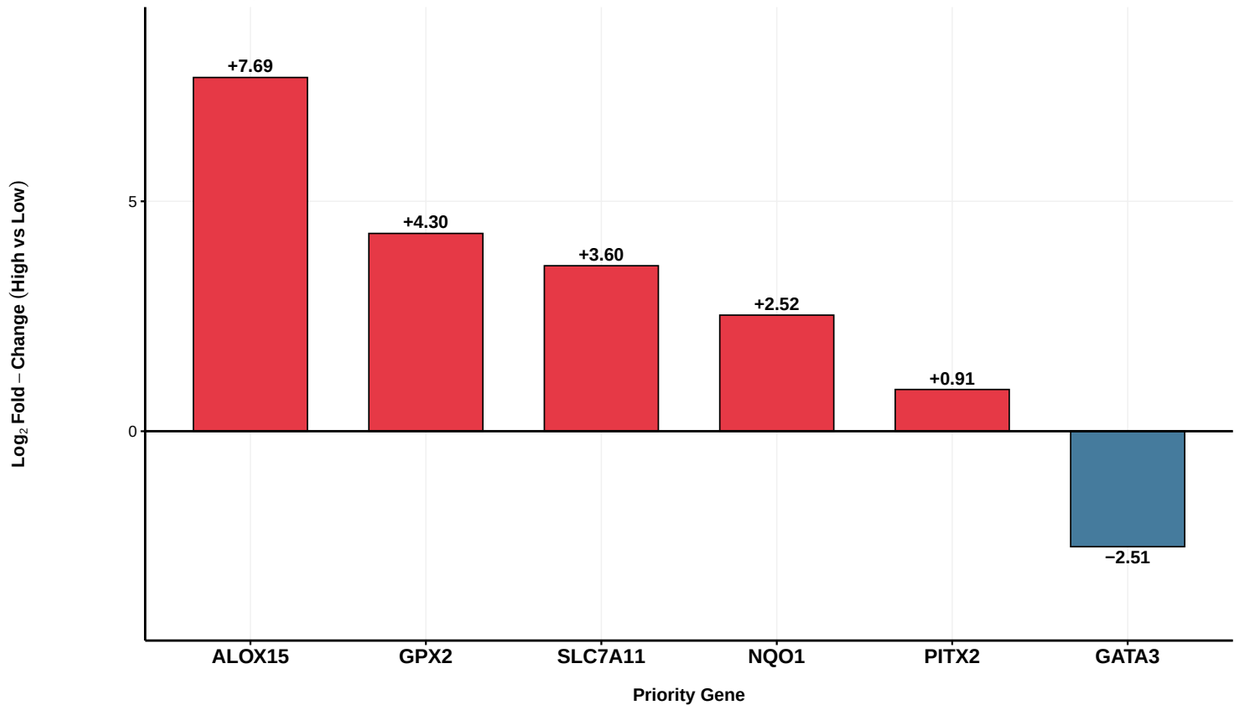
D Metastatic Niche Amplification

Primary vs tLN Malignant (P = 1.78e−26, Cliff's d = +0.26)



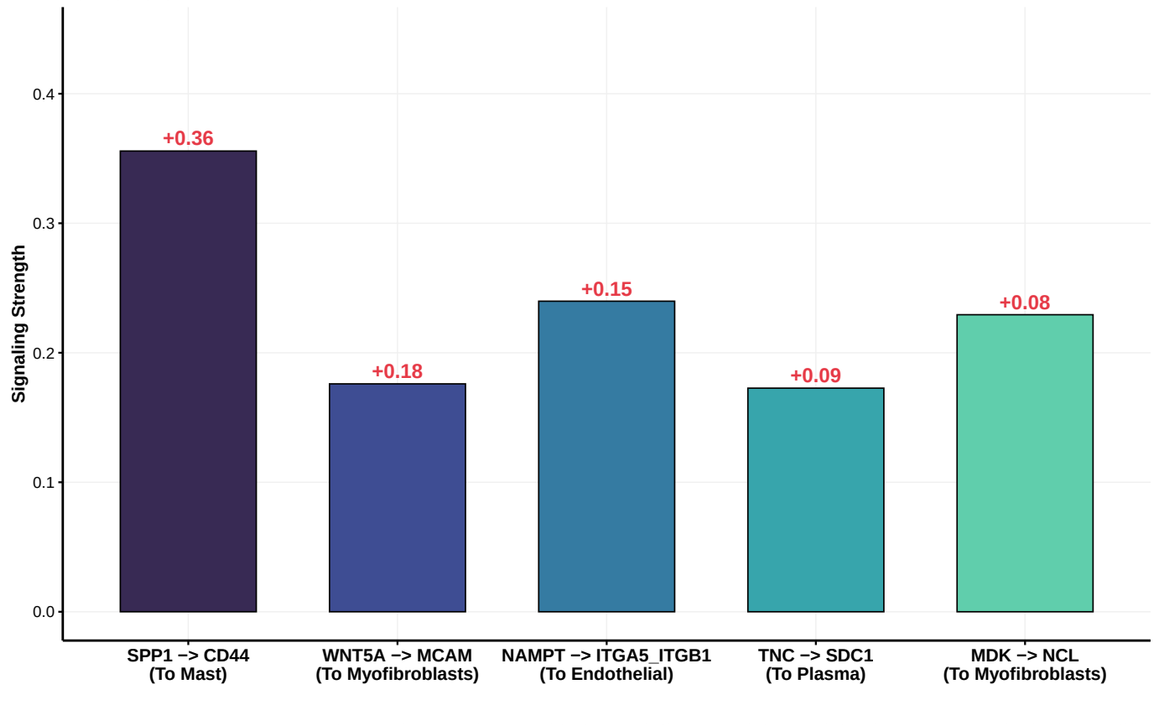
E Priority Gene Single−Cell Divergence (NRF2−High vs Low)

Transcriptomic ground truth for Phase 1−3 discoveries



F NRF2−High Intercellular Communication Rewiring

Top outgoing ligand−receptor axes (labeled with delta vs NRF2−Low)



Paired single−cell ground truth resolution of bulk transcriptomic and immunological desertification phenomena